Optimistic rollup. A scaling solution whereby transactions are aggregated off-chain into a single *digest* that is submitted to the chain on a periodic basis.

Oracle. A method whereby information is gathered outside of a *blockchain*. Parties must agree on the source of the information.

Order book matching. A process in which all parties must agree on the swap exchange rate. Market makers can post bids and asks to a *decentralized exchange (DEX)* and allow takers to fill the quotes at the pre-agreed price. Until the offer is taken, the market maker has the right to withdraw the offer or update the exchange rate.

Perpetual futures contract. Similar to a traditional futures contract but without an expiration date.

Proof of stake (PoS). An alternative consensus mechanism, and a key feature of Ethereum 2.0, in which the staking of an asset on the next block replaces the mining of blocks as in *proof of work (PoW)*. In PoW, miners need to spend on electricity and equipment to win a block. In proof of stake, validators commit some capital (the stake) to attest that the block is valid. Validators make themselves available by staking their cryptocurrency, and then they are randomly selected to propose a block. The proposed block needs to be attested by a majority of the other validators. Validators profit by both proposing a block and attesting to the validity of others' proposed blocks. If validators act maliciously, there is a penalty mechanism whereby their stake is *slashed*.